
Online Examination Report

Date:	14.08.2014
Attempts:	1
Examinee:	Mwakagile, Faraja
Date of birth:	7/27/1979
Subject:	080- Principles of flight
Result:	37,5 %
Time used:	00:09:32 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	222	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	201	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
3	219	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	223	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	199	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
6	187	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
7	192	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	190	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	229	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	147	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
11	200	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
12	184	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
13	185	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	113	81.1	SUBSONIC AERODYNAMICS	1,00	1,00
15	48	81.1.1.1	Laws and definitions	0,00	1,00
16	20	81.1.1.1	Laws and definitions	1,00	1,00
17	25	81.1.1.1	Laws and definitions	0,00	1,00
18	71	81.1.1.1	Laws and definitions	0,00	1,00
19	37	81.1.1.3	Aerodynamic forces and moments on aerofoils	1,00	1,00
20	61	81.1.1.4	Shape of an aerofoil section	0,00	1,00
21	17	81.1.1.5	Wing shape	0,00	1,00
22	30	81.1.2	Two-dimensional airflow around an aerofoil	0,00	1,00
23	27	81.1.2.2	Stagnation point	1,00	1,00
24	84	81.1.3.1	The lift coefficient C_l	1,00	1,00
25	90	81.1.4.2	Induced drag	0,00	1,00
26	54	81.1.4.2	Induced drag	1,00	1,00
27	74	81.1.4.2	Induced drag	0,00	1,00
28	38	81.1.8.2	The stall speed	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	28	81.1.8.5	Special phenomena of stall	0,00	1,00
30	78	81.1.12.1	Ice and other contaminants	0,00	1,00
31	18	81.2.3.2	Effect on lift	0,00	1,00
32	39	81.2.3.4	Effect on pitching moment	0,00	1,00
33	45	81.2.4	Buffet onset	1,00	1,00
34	72	81.5.4.5	Means to avoid adverse yaw:	0,00	1,00
35	77	81.5.8.3	Stabiliser trim	1,00	1,00
36	32	81.5.8.3	Stabiliser trim	1,00	1,00
37	108	81.7	PROPELLERS	1,00	1,00
38	76	81.7.1.1	Relevant propeller parameters	0,00	1,00
39	24	81.8.1.2	Straight steady climb	0,00	1,00
40	29	81.8.1.4	Straight steady glide	0,00	1,00
Total:38%				15,00	40,00

Attachment 2: List of result by topic

Licence

Date: 13.08.2014

Examinee: Mwakagile, Faraja

Location: Uganda Civil Aviation Authority

Your Result: 15,00 from 40,00 Points (37,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	15,00	40,00	37,50
81.1. SUBSONIC AERODYNAMICS	17	6,00	17,00	35,29
81.2. HIGH SPEED AERODYNAMICS	3	1,00	3,00	33,33
81.5. CONTROL	3	2,00	3,00	66,67
81.7. PROPELLERS	2	1,00	2,00	50,00
81.8. FLIGHT MECHANICS	2	0,00	2,00	0,00
Total	40	15,00	40,00	37,50

Candidate: Faraja Mwakagire
Date of birth: 1979-07-27

Subject: 080- Principles of flight
Date: 2014-08-13

1

Which stall must be performed during a flight instructor - airplane practical test?

☐ Power-on or power-off.

☐ Imminent.

☒ Accelerated.

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2

A rapid acceleration can create the illusion of being in a

- ☒ noseup attitude.
- ☐ nosedown attitude.
- ☐ left turn.

3

How should a roundout from a moderate-rate ascent to level flight be made?

- ☐ Vent at altitude and add heat upon settling back down to altitude.
- ☒ Cool the envelope by venting and add heat just before arriving at the desired altitude.
- ☐ Reduce the amount of heat gradually as the balloon approaches altitude.

4

Which is the correct symbol for the stalling speed or the minimum steady flight speed at which the airplane is controllable?

☒ $V(S1)$.

☐ $V(S)$.

☐ $V(S0)$.

5

The best lift/drag ratio of a glider is a value that

- ☒ varies depending upon the weight being carried.
- ☐ remains constant and is independent of the weight being carried.
- ☐ remains constant regardless of airspeed changes.

6

Which characteristic of a spin is not a characteristic of a steep spiral?

- ☐ Rapid loss of altitude.
- ☐ High rate of rotation.
- ☒ Stalled wing.

7

An aircraft is loaded with the CG aft of the aft limit. What effect will this have on controllability?

- ☐ Stall and spin recovery may be difficult or impossible.
- ☒ A stall will occur at a higher indicated airspeed due to the greater downloading on the elevator.
- ☐ A stall will occur at a lower airspeed, but recovery will be easier because of reduced wing loading.

8

Maximum gliding distance of an aircraft is obtained when

- ☐ induced drag and parasite drag are equal.
- ☒ induced drag equals the coefficient of lift.
- ☐ parasite drag is the least.

9

How does the wing design feature 'washout' affect the production of lift?

- ☐ The center of lift moves from the trailing edge of the wing, to the leading edge of the wing, as the wing begins to stall.
- ☒ The main body of the wing continues to produce lift when the wing tips are not producing lift.
- ☐ The wing tips continue producing lift when the main body of the wing is not producing lift.

10

When a slight upward or negative flap deflection is used, the result is

☐ decreased lift.

☒ decreased drag.

☐ increased drag.

11

In comparison to an approach in a moderate headwind, which is an indication of a possible wind shear due to a decreasing headwind when descending on the glide slope?

- ☒ Higher pitch attitude is required.
- ☐ Less power is required.
- ☐ Lower descent rate is required.

12

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☐ have no effect on the CG location.
- ☒ cause the CG location to move aft.
- ☐ cause the CG location to move forward.

13

If the landing gear on an airplane moves forward during retraction, the total moment will

☐ decrease.

☒ remain the same.

☐ increase.

14

When the angle of attack of a symmetrical airfoil is increased, the center of pressure will

- ☐ remain unaffected.
- ☐ move aft along the airfoil surface.
- ☒ have very little movement.

15

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The static pressure in the undisturbed airflow is lower than in the throat.
II. The speed in the undisturbed airflow is lower than in the throat.

- ☐ I is correct, II is incorrect.
- ☒ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.

16

The SI unit of measurement for density is:

☒ **kg/m³.**

☐ **kg/cm².**

☐ **bar.**

☐ **psi.**

17

Which of these statements about a stationary subsonic flow are correct or incorrect?

- I. The static pressure decreases as the streamlines converge.
II. The velocity increases as the streamlines converge.

- ☒ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.

18

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The dynamic pressure in the throat is lower than in the undisturbed airflow.
II. The total pressure in the throat is lower than in the undisturbed airflow.

- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is incorrect.

19

The angle between the direction of the undisturbed airflow (relative wind) and the chord line of an aerofoil is called:

☐ same as the angle between chord line and fuselage axis.

☒ angle of attack.

☐ climb path angle.

☐ glide path angle.

20

An aerofoil is cambered when:

- ☐ the chord line is curved.
- ☐ the maximum thickness is large compared with the length of the chord.
- ☒ the upper surface of the aerofoil is curved.
- ☐ the line, which connects the centres of all inscribed circles, is curved.

21

Taper ratio of a wing is the ratio between:

- ☐ wing span squared and wing area.
- ☒ tip chord and root chord.
- ☐ mean geometric chord and wing span.
- ☒ root chord and tip chord.

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22

(Please view annex 081-0005 issue date July 2004)

How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

- ☐ $V_1 = 0$ and $V_2 > V$.
- ☐ $V_1 = 0$ and $V_2 = V$.
- ☐ $V_1 > V_2$ and $V_2 < V$.
- ☒ $V_1 < V_2$ and $V_2 < V$.

See Attachments:

1. 081-0005.gif

Annex

(Attachment
No.1)

23

The stagnation point is the point:

- ☐ of the intersection of the total aerodynamic force and the chord line.
- ☐ relative to which the sum of all moments is independent of angle of attack.
- ☐ of the intersection of the thrust vector and the chord line.
- ☒ where the velocity of the relative airflow is reduced to zero.

24

The lift formula can be written as:
(rho = density)

☐ $L = CL \cdot 2\rho \cdot V^2 \cdot S.$

☐ $L = W.$

☒ $L = CL \cdot \frac{1}{2}\rho \cdot V^2 \cdot S.$

☐ $L = n \cdot W.$

25

Excluding constants, the coefficient of induced drag (C_{Di}) is the ratio of:

- ☐ CL^2 and AR (aspect ratio).
- ☒ CL and b (wing span).
- ☐ CL^2 and S (wing surface).
- ☐ CL and C_D .

26

Winglets:

- ☐ increase the manoeuvrability.
- ☐ decrease the static lateral stability.
- ☒ decrease the induced drag.
- ☐ create an elliptical lift distribution.

27

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☐ lift increases by a factor of 4.
- ☒ angle of attack increases by a factor of 1/4.
- ☐ induced drag increases by a factor of 4.
- ☐ total drag increases by a factor of 4.

28

The stall speed increases, when: (all other factors of importance being constant)

- ☒ weight decreases.
- ☐ spoilers are retracted.
- ☐ pulling out of a dive.
- ☐ minor altitude changes occur e.g. 0-10.000 ft.

29

A stick pusher:

- ☐ pushes the elevator control forward prior to stick shaker activation.
- ☒ pushes the elevator control to avoid a stall at a negative load factor.
- ☐ pushes the elevator control forward when a specified value of angle of attack is exceeded.
- ☐ vibrates the elevator control.

30

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

- ☐ decrease of CLMAX and decrease of drag.
- ☒ increase of CLMAX and decrease of drag.
- ☐ decrease of CLMAX and increase of drag.
- ☐ increase of CLMAX and increase of drag.

31

During which type of stall does the angle of attack have the smallest value?

☒ Accelerated stall.

☐ Deep stall.

☐ Shock stall.

☐ Low speed stall.

32

Which of these statements about "tuck under" are correct or incorrect?

- I. A contributing factor to "tuck under" is a forward movement of the centre of pressure of the wing.
- II. A contributing factor to "tuck under" is an increase in the downwash angle at the location of the horizontal stabiliser.

- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.

33

What is the significance of the maximum allowed cruising altitude, based on the 1.3 g margin?

- ☐ a manoeuvre with a load factor of 1.3 will cause a Mach number at which accelerated low speed stall occurs.
- ☐ a manoeuvre with a load factor of 1.3 will cause Mcrit to be exceeded.
- ☒ a manoeuvre with a load factor of 1.3 will cause buffet onset.
- ☐ exceeding a load factor of 1.3 will cause permanent deformation of this aeroplane.

34

An example of differential aileron deflection during initiation of left turn is:

- ☐ Left aileron: 2° down.
Right aileron: 5° up.
- ☐ Left aileron: 5° up.
Right aileron: 2° down.
- ☒ Left aileron: 5° down.
Right aileron: 2° up.
- ☐ Left aileron: 2° up.
Right aileron: 5° down.

35

The most important factor determining the required position of the Trimmable Horizontal Stabiliser (THS) for take off is the:

- ☐ stall speed.
- ☒ position of the aeroplane's centre of gravity.
- ☐ centre of gravity position of the fuel.
- ☐ total mass of the aeroplane.

36

When comparing a stabiliser trim system with an elevator trim system, which of these statements is correct?

- ☐ an elevator trim is able to compensate larger changes in pitching moments.
- ☒ a stabiliser trim is able to compensate larger changes in pitching moments.
- ☐ an elevator trim is more suitable for aeroplanes with a large CG range.
- ☐ a stabiliser trim is more sensitive to flutter.

37

Blade angle of a propeller is defined as the angle between the

- ☐ angle of attack and chord line.
- ☐ angle of attack and line of thrust.
- ☒ chord line and plane of rotation.

38

The difference between a propeller's blade angle and its angle of attack is called:

- ☐ the effective pitch.
- ☐ the propeller angle.
- ☒ propeller slip.
- ☐ the helix angle.

39

An aeroplane climbs to cruising level with a constant pitch attitude and maximum climb thrust, (assume no supercharger).
How do the following variables change during the climb? (Gamma = flight path angle)

- ☒ Gamma decreases, angle of attack increases, IAS remains constant.
- ☐ Gamma remains constant, angle of attack remains constant, IAS decreases.
- ☐ Gamma decreases, angle of attack increases, IAS decreases.
- ☐ Gamma decreases, angle of attack remains constant, IAS decreases.

40

Which of the following increases the maximum duration of a glide?

- ☐ A headwind.
- ☐ An increase in mass.
- ☐ A decrease in mass.
- ☒ A tailwind.

Attachment No. 1

Questions 22

Annex

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081-0005

